



# NatureVeda

#Medicinal Plant Identification and  
Remedy finder Platform

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Student Name:  
**Vibhor Singh**  
**Priyansh Singhal**  
**Raghvendra Dwiwedi**  
**Avanish Mishra**

Guide:  
**Ankit Dixit Sir**



BY  
**TEAM ABHEDYA**

# OUTLINE

- Abstract of the Project
- Problem Statement
- Proposed Solution
- System Architecture
- Live Demo of the Project
- Conclusion & Future Scope

# ABSTRACT

**NatureVeda** is a full-stack **AI-powered wellness platform** that bridges traditional **Ayurvedic** and **Homeopathic** medicine with modern technology. It lets users **identify medicinal plants from photos** using a **trained machine-learning model**, discover their unique **Ayurvedic constitution (dosha)** through a **20-question assessment**, **describe symptoms in English, Hindi, or Hinglish** to receive traditional **remedy guidance**, and **scan lab reports** to automatically flag out-of-range values with **matching Ayurvedic and Homeopathic suggestions**. The platform combines a **Next.js web frontend**, a **FastAPI backend**, a **custom-trained MobileNetV2 plant classifier**, and **OCR-based report analysis** — all built to run on free-tier infrastructure.

# PROBLEM STATEMENT

- **Traditional Ayurvedic and Homeopathic knowledge** is scattered, hard to access, and rarely presented in a structured, searchable digital format.
- **Most people can't identify medicinal plants** by sight or know which traditional remedies apply to their symptoms.
- **Lab reports are full of jargon and reference ranges that are confusing to interpret without a doctor**, and traditional wellness guidance is rarely connected back to actual report values.
- Existing AYUSH-related apps are mostly static content libraries — none combine **AI-based plant recognition, personalized dosha assessment, symptom-based matching, and report analysis** in a single connected experience.

# PROPOSED SOLUTION

## **NatureVeda addresses this with five integrated features:**

- **Plant Scanner** — upload a photo, get instant species identification via a custom-trained image classifier, plus Ayurvedic details (dosha effect, preparation, dosage, safety).
- **Dosha Assessment** — a 20-question quiz scores users across Vata, Pitta, and Kapha and generates a personalized diet, routine, and exercise plan.
- **Symptom Checker** — free-text symptom input (multilingual, including Hinglish) matched via TF-IDF similarity against a curated symptom-remedy dataset, returning both Ayurvedic and Homeopathic suggestions.
- **Report Scanner** — OCRs an uploaded lab report, parses test values against reference ranges, flags high/low results, and suggests relevant traditional remedies for each flagged value.
- **Explore Plants / Therapies / Homeopathy libraries** — full searchable databases for independent browsing.

# Live Demo

**VISIT US AT**

<https://nature-veda.vercel.app>

**OR**



# CONCLUSION & FUTURE SCOPE

## Conclusion:

**NatureVeda** demonstrates that **traditional Ayurvedic and Homeopathic** wellness knowledge can be made **accessible, personalized, and interactive** through AI — without requiring paid infrastructure, **using free-tier hosting, open datasets, and lightweight ML models** suited to those constraints.

# CONCLUSION & FUTURE SCOPE

## Future Scope:

- **Expand the plant classifier dataset for higher accuracy across more species**
- **Add DOCX/handwritten report parsing**
- **Enable authenticated user accounts to save scan/assessment history long-term (Supabase auth is already wired in)**
- **Add SMS/phone-based sign-in once budget allows**
- **Extend to a dedicated mobile app**
- **Expand the homeopathy and symptom datasets for broader coverage**



# SYSTEM ARCHITECTURE

## Frontend



Next.js (web)— fully responsive web app (mobile, tablet, desktop)

## Backend



FastAPI (Python) — REST APIs for plants, symptoms, dosha, reports

## ML



MobileNetV2 (TFLite) for plant ID; TF-IDF for symptom matching

## OCR



Hosted OCR.space API for report scanning

## Auth/DB



Supabase

**Hosting** —————> Vercel (frontend), Render (backend)

**Thank You**